

EE 210L: Digital Circuits and Logic Design Lab

Lab 5: NAND and NOR as Universal Gates

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels only, 0 or 1. Logic diagrams stay on your Canvas pre-lab; carry only expressions and counts across. The Lab 4 columns come from your own Lab 4 report.

Part 4: Carried Over from the Pre-Lab

Function	Expression
W, minimum SOP from Lab 4	
W, as a nest of NAND operations	
P, the form you built in Lab 4	
P, as a nest of NAND operations	

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: NOT, AND, and OR from NAND Alone

Gates used: NOT _____ AND _____ OR _____

a	Predicted a'	Measured a'
0		
1		

a	b	Predicted a·b	Measured a·b	Predicted a+b	Measured a+b
0	0				
0	1				
1	0				
1	1				

Do all three tables match the 74LS04, 74LS08, and 74LS32 results from Lab 1?

Part 5.3: NOT, OR, and AND from NOR Alone

Gates used: NOT _____ OR _____ AND _____

a	Predicted a'	Measured a'
0		
1		

a	b	Predicted a+b	Measured a+b	Predicted a·b	Measured a·b
0	0				
0	1				
1	0				
1	1				

NAND reached AND in two gates and OR in three. NOR did the opposite. Which form, sum-of-products or its dual, does each gate suit?

Part 5.4: Function W, Built from NAND Only

NAND gates used _____ 74LS00 packages _____

#	a	b	c	d	Predicted W	Measured W, all-NAND	Measured W, Lab 4
1	0	0	0	0			
2	0	0	1	0			
3	0	0	1	1			
4	0	1	0	0			
5	0	1	0	1			
6	0	1	1	0			
7	0	1	1	1			
8	1	0	0	0			
9	1	0	1	0			
10	1	0	1	1			
11	1	1	0	1			
12	1	1	1	0			

All twelve rows agree across two circuits with different gates and different chips. State what that shows about the conversion in §3.4.

Part 5.5: Function P, Built from NAND Only

NAND gates used _____ of the four on the 74LS00 Chips the Lab 4 version needed _____

#	Digit	a	b	c	d	Predicted P	Measured P, all-NAND	Measured P, Lab 4
1	0							
2	1							
3	2							
4	3							
5	4							
6	5							
7	6							
8	7							

Part 5.6: Cost Comparison

Count 2-input gates. Count inverters as gates here, since a NAND inverter is one of the four on the package.

Function	Version	Gates	Packages	Part numbers used
W	Lab 4, AND-OR			
W	All-NAND			
P	Lab 4, AND-OR-NOT			
P	All-NAND			

Which version uses more gates, and which uses more packages? Which of the two numbers would decide the cost of a thousand boards?

Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: what makes a gate universal, why the SOP to NAND conversion leaves the wiring unchanged, and why a design built from one part number can be cheaper

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than the same design built from three even when it uses more gates.
